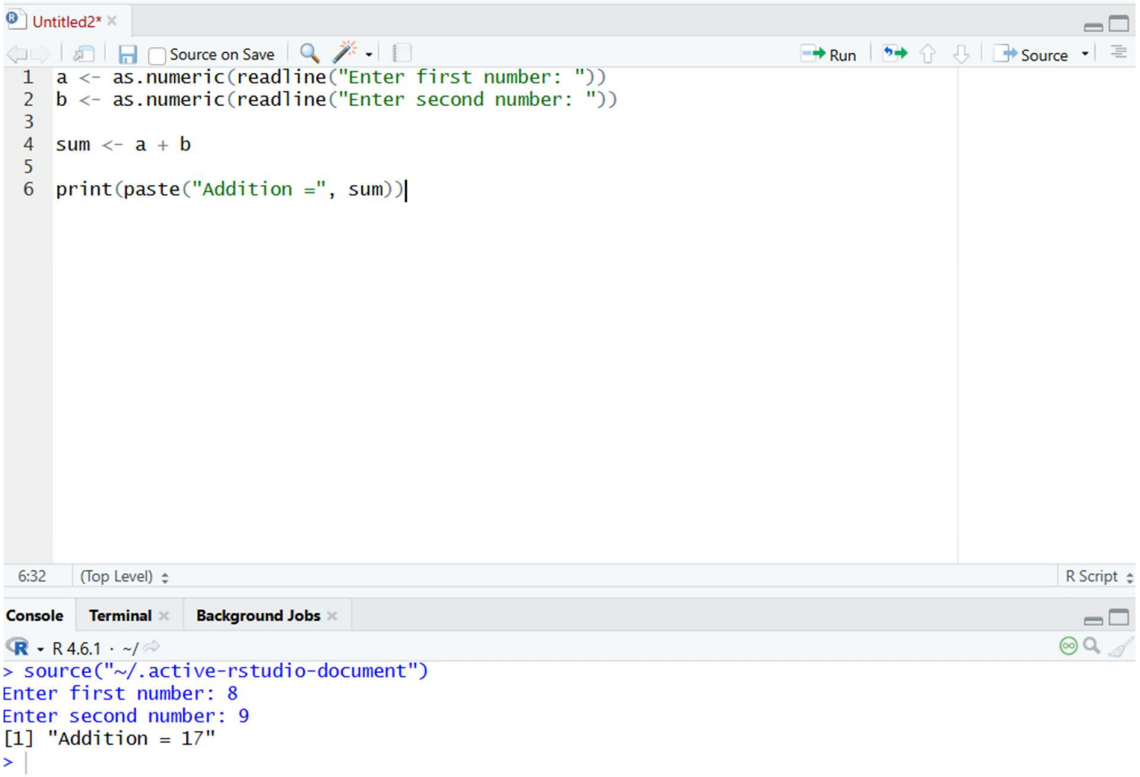
R-Programming

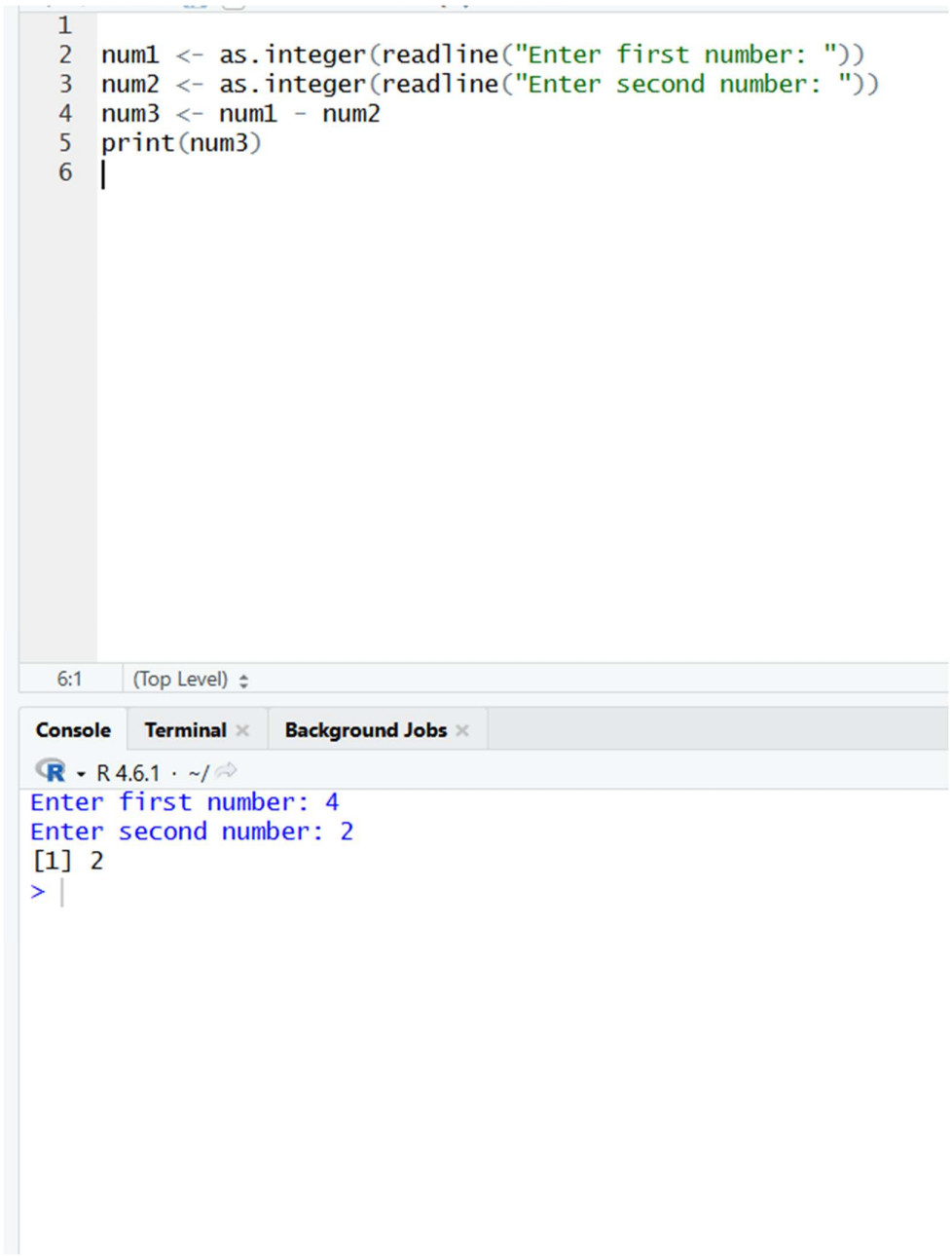
Exp-1:



Result:

Thus the basic program addition is executed successfully.

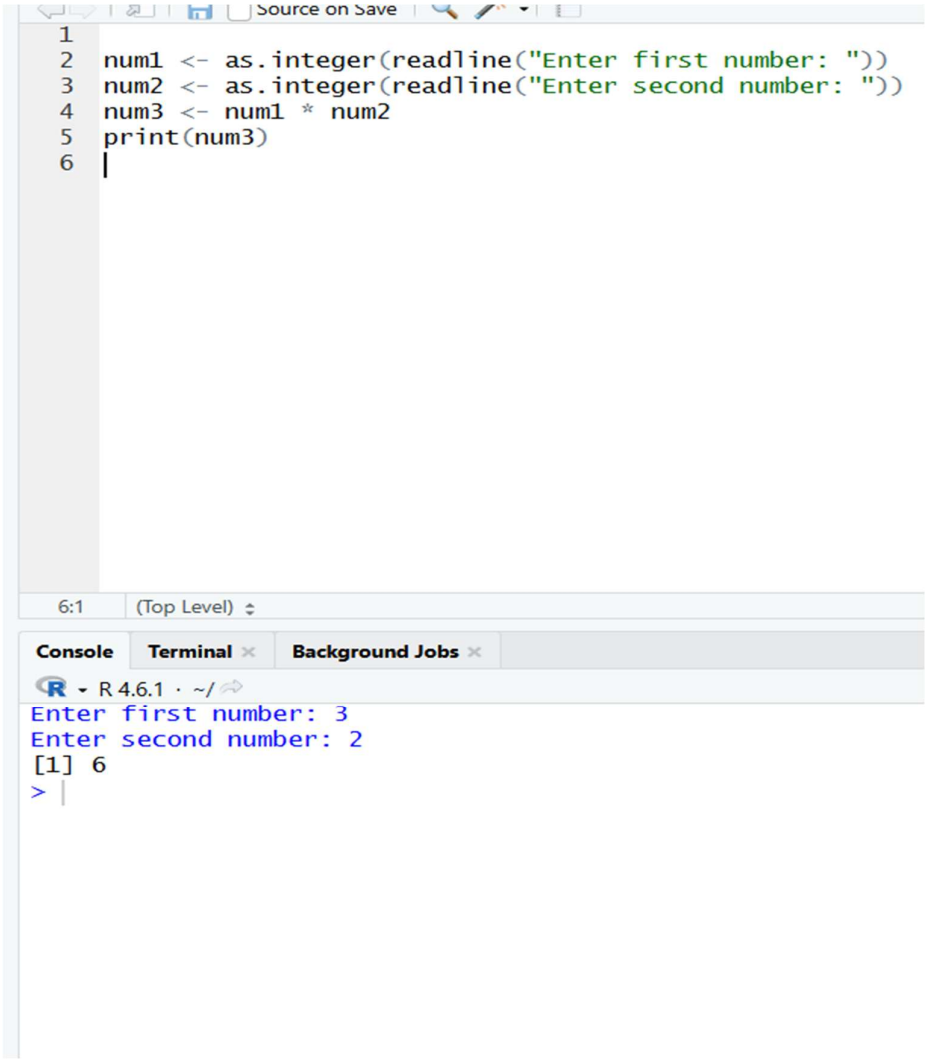
Exp-2:



Result:

Thus, the basic program subtraction is executed successfully.

Exp-3:



Result:

Thus, the basic program multiplication is executed successfully.

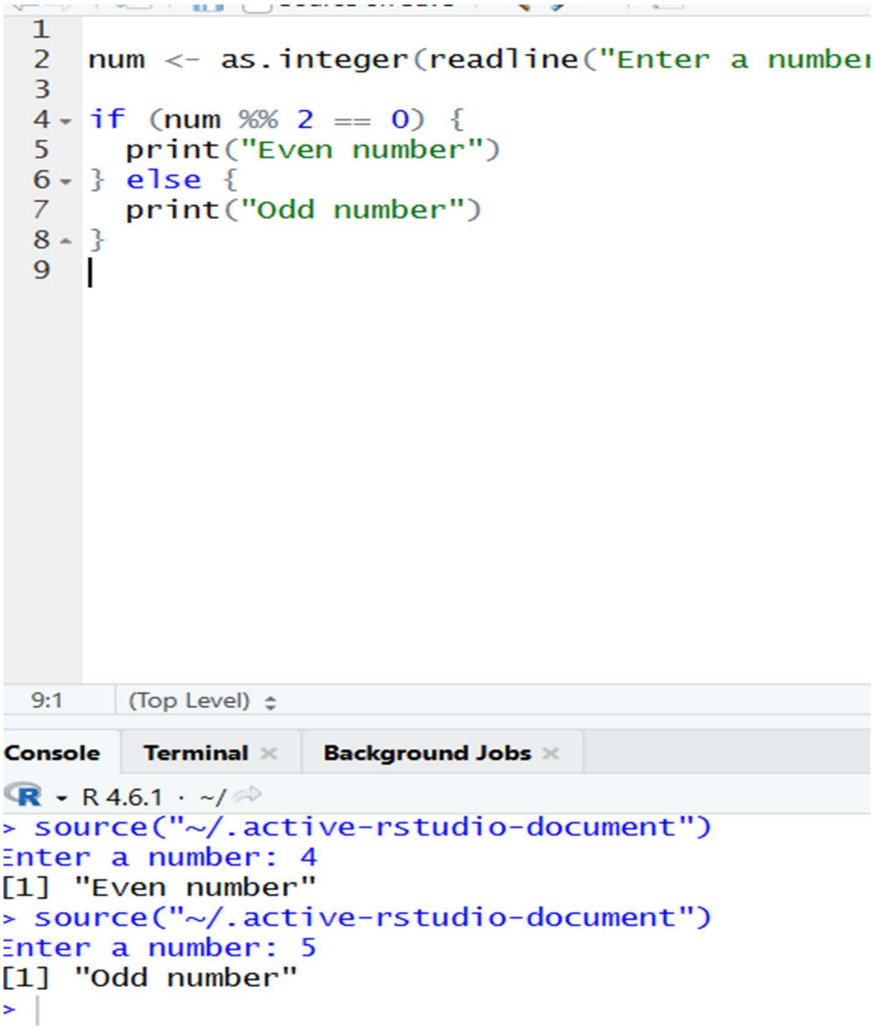
Exp-4:



Result:

Thus, the basic program division is executed successfully.

Exp-5:



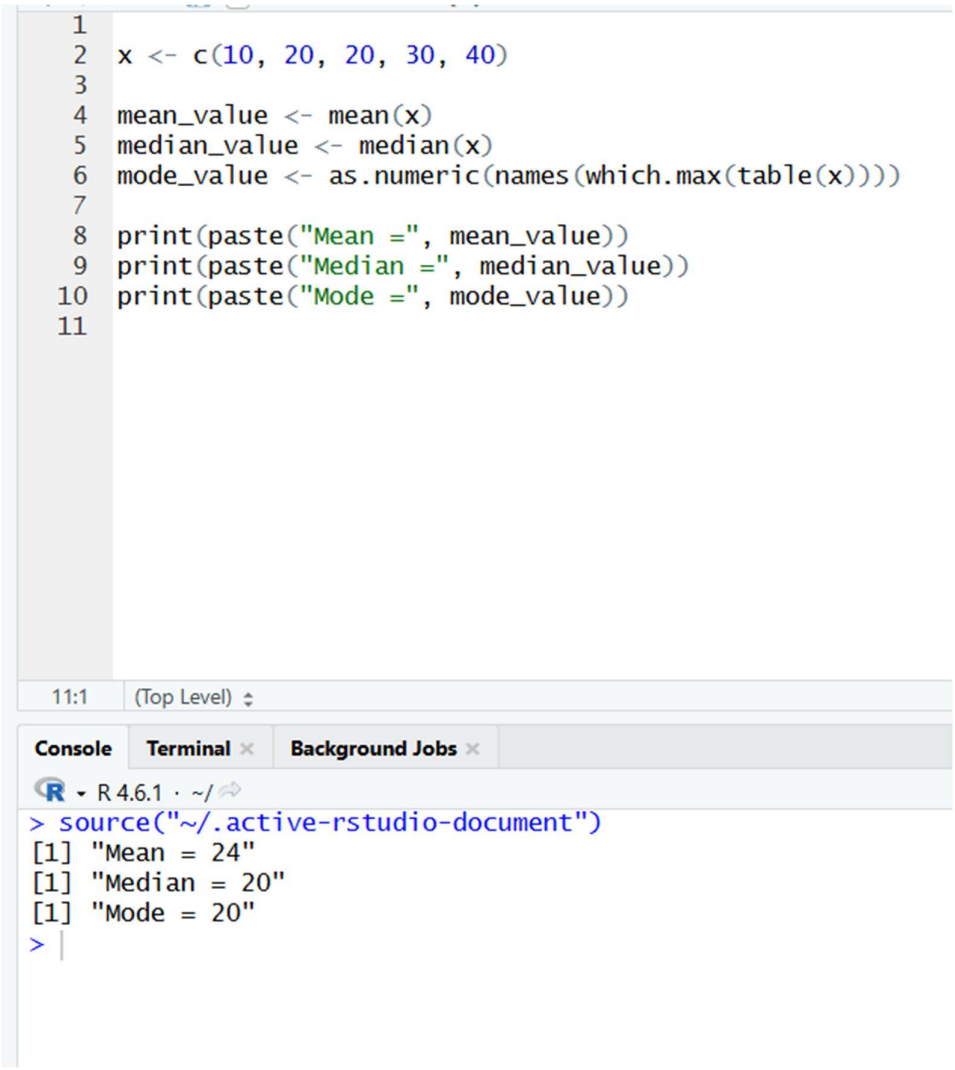
```
num <- as.integer(readline("Enter a numbe
if (num %% 2 == 0) {
  print("Even number")
} else {
  print("Odd number")
}
```

```
9:1  (Top Level)
Console  Terminal  Background Jobs
R 4.6.1 · ~/
> source("~/.active-rstudio-document")
Enter a number: 4
[1] "Even number"
> source("~/.active-rstudio-document")
Enter a number: 5
[1] "Odd number"
>
```

Result:

Thus, the basic program odd or even is executed successfully.

Exp-6:



```r

x <- c(10, 20, 20, 30, 40)

mean_value <- mean(x)
median_value <- median(x)
mode_value <- as.numeric(names(which.max(table(x))))

print(paste("Mean =", mean_value))
print(paste("Median =", median_value))
print(paste("Mode =", mode_value))

```

11:1 (Top Level)

Console Terminal Background Jobs

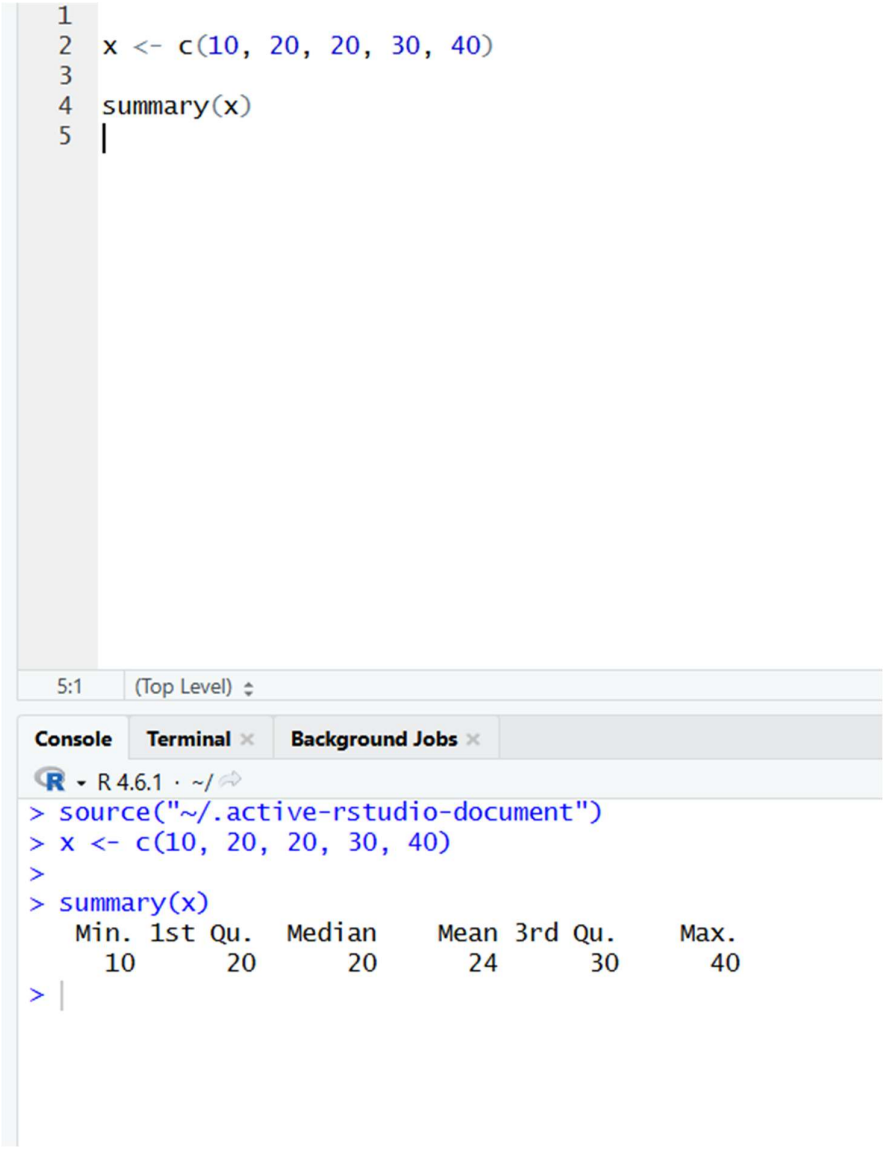
R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
[1] "Mean = 24"
[1] "Median = 20"
[1] "Mode = 20"
>
```

Result:

Thus the central tendency and measure of dispersion is executed successfully.

Exp-7:



Result:

Thus, the central tendancy and measure of dispersion is executed successfully.

Exp-8:

```r
a <- as.numeric(readline("Enter first number: "))
b <- as.numeric(readline("Enter second number: "))
c <- as.numeric(readline("Enter third number: "))

if (a >= b && a >= c) {
  print(a)
} else if (b >= a && b >= c) {
  print(b)
} else {
  print(c)
}
```

13:1 (Top Level)

Console Terminal Background Jobs

R 4.6.1 · ~/

```
Enter first number: 10
Enter second number: 40
Enter third number: 30
[1] 40
>
```

Result:

Thus, the greatest among the three numbers was executed successfully.

Exp:9

```
1
2  x <- c(10, 20, 20, 30, 40, 50, 60)
3
4  q1 <- quantile(x, 0.25)
5  q3 <- quantile(x, 0.75)
6  iqr_value <- IQR(x)
7
8  print(paste("Q1 =", q1))
9  print(paste("Q3 =", q3))
10 print(paste("IQR =", iqr_value))
11
```

10:33 (Top Level)

Console Terminal Background Jobs

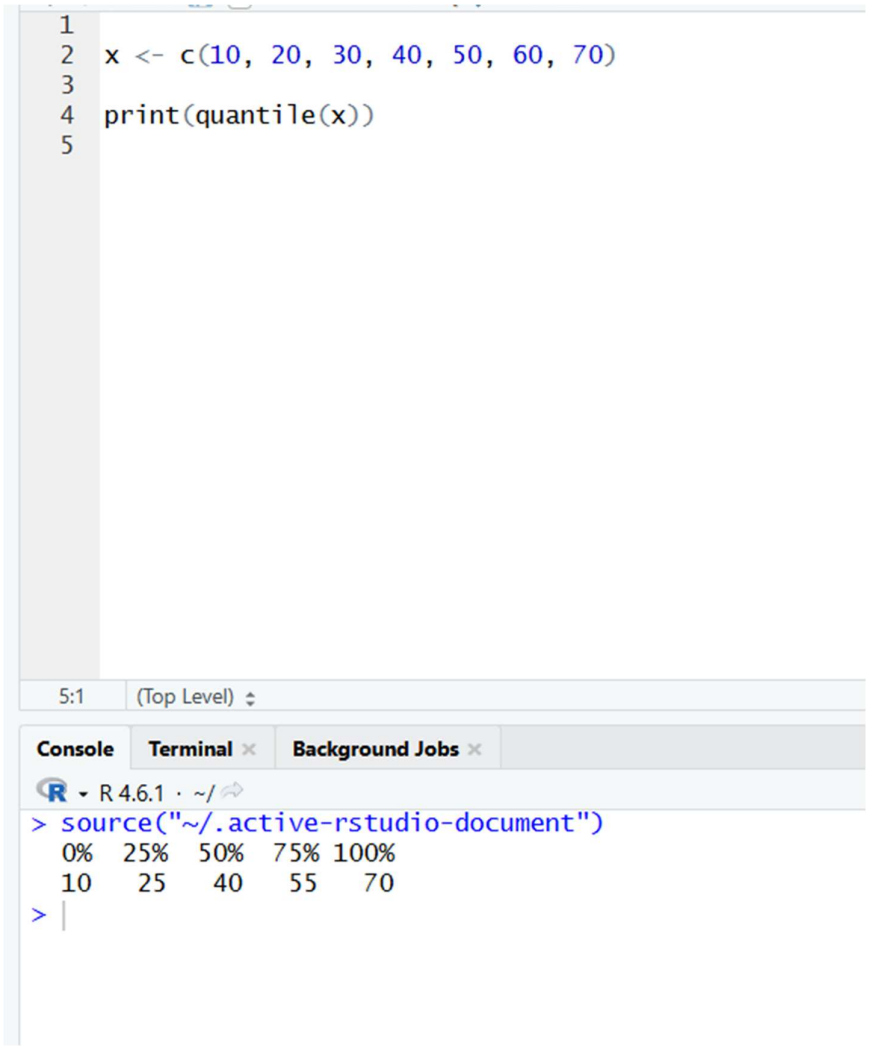
R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
[1] "Q1 = 20"
[1] "Q3 = 45"
[1] "IQR = 25"
>
```

Result:

Thus, the program for central tendency and data dispersion measures was executed successfully.

Exp-10



Result:

Thus, the program for central tendency and data dispersion measures was executed successfully

Exp-11:

```
1
2  x <- c(10, 20, 30, 40, 50)
3
4  mid_range <- (min(x) + max(x)) / 2
5
6  print(paste("Mid-Range =", mid_range))
7
```

7:1 (Top Level)

Console Terminal Background Jobs

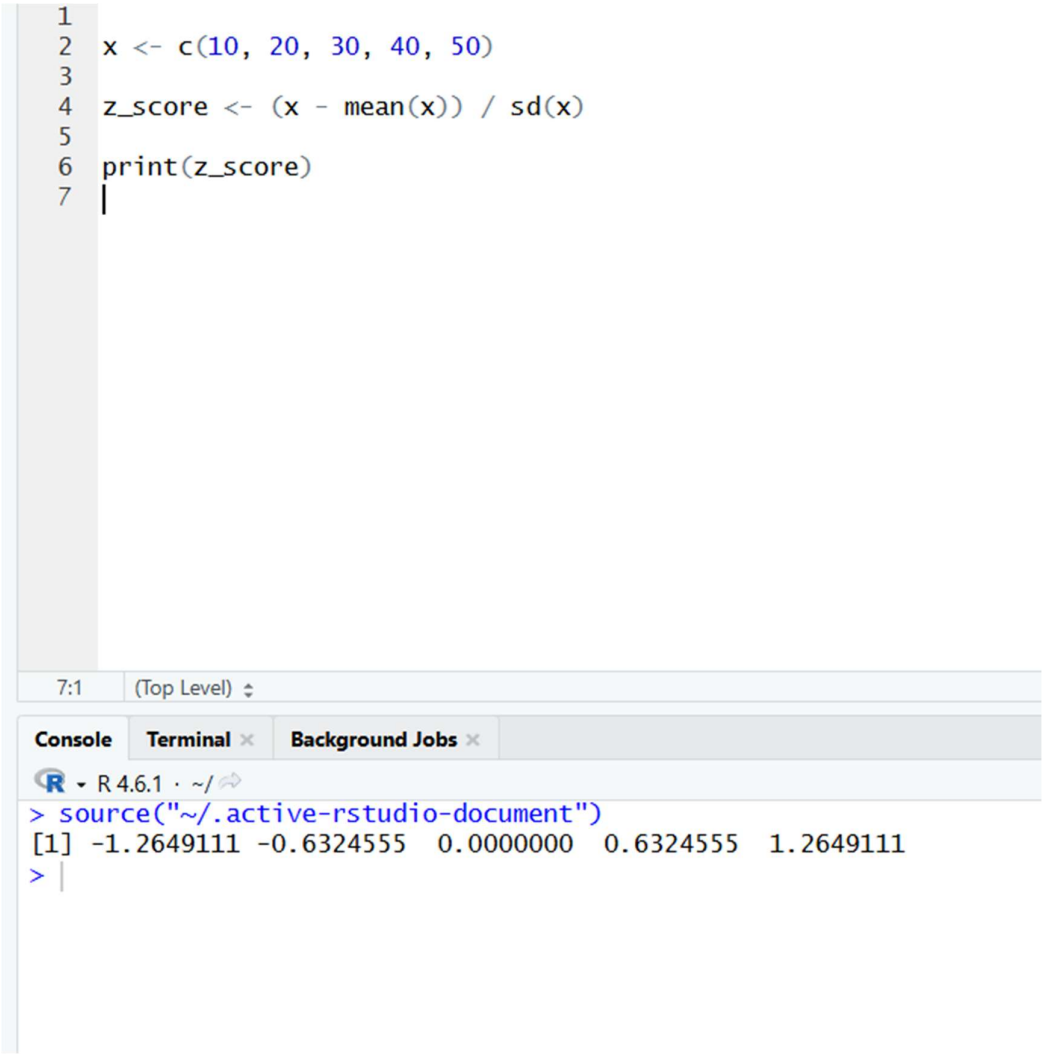
R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
[1] "Mid-Range = 30"
>
```

Result:

Thus, the program for central tendency and data dispersion measures was executed successfully.

Exp-12:



Result:

Thus, the Z-score normalization using R tool was executed successfully.

Exp-13:

```
1
2  x <- c(10, 20, 30, 40, 50)
3
4  min_value <- min(x)
5  max_value <- max(x)
6  mean_value <- mean(x)
7
8  minmax <- (x - min_value) / (max_value - min_value)
9
10 print(paste("Minimum =", min_value))
11 print(paste("Maximum =", max_value))
12 print(paste("Mean =", mean_value))
13 print("Min-Max Normalization:")
14 print(minmax)
15
```

14:14 (Top Level)

Console Terminal Background Jobs

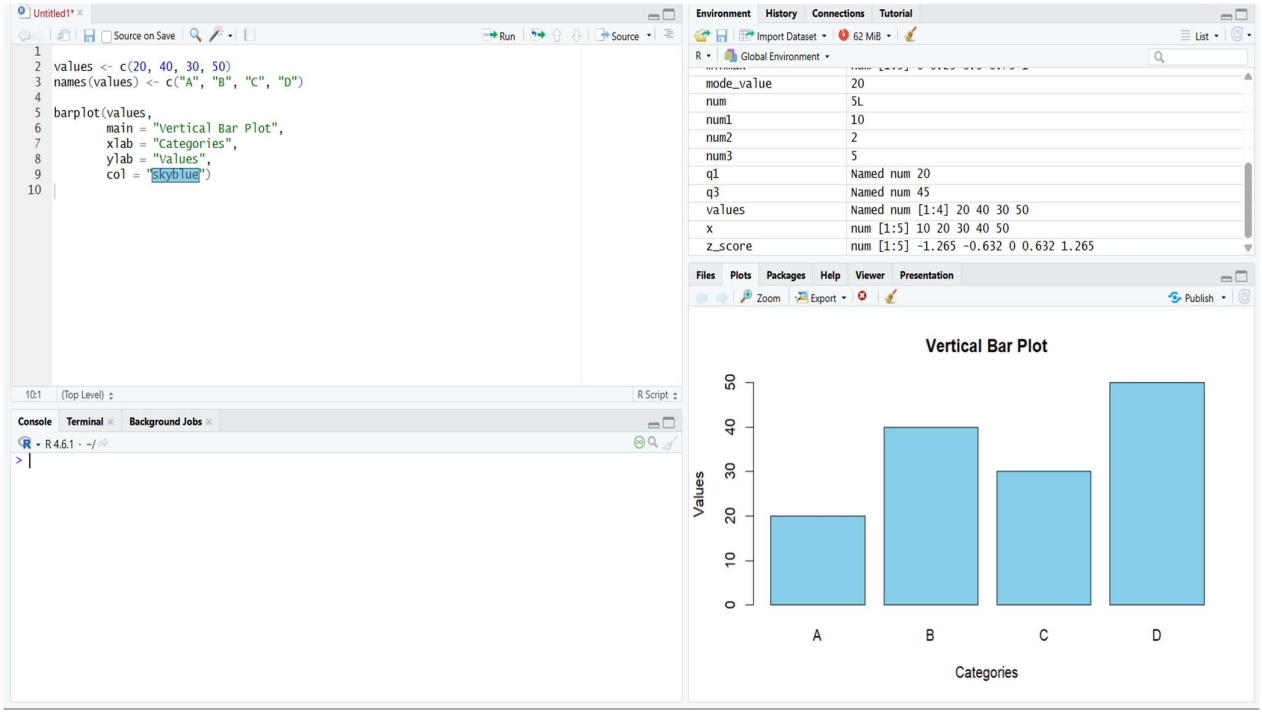
R 4.6.1 · ~/

```
> source("~/.active-rstudio-document")
[1] "Minimum = 10"
[1] "Maximum = 50"
[1] "Mean = 30"
[1] "Min-Max Normalization:"
[1] 0.00 0.25 0.50 0.75 1.00
>
```

Result:

Thus, the program for min,max,minmax,mean was executed successfully.

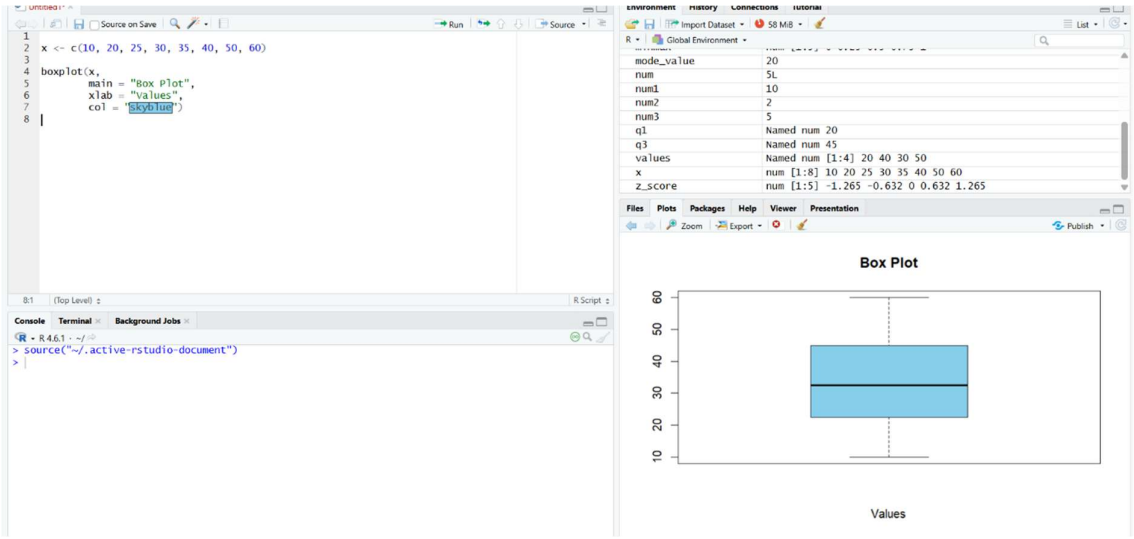
Exp-14:



Result:

Thus, the bar and horizontal bar plot was executed successfully.

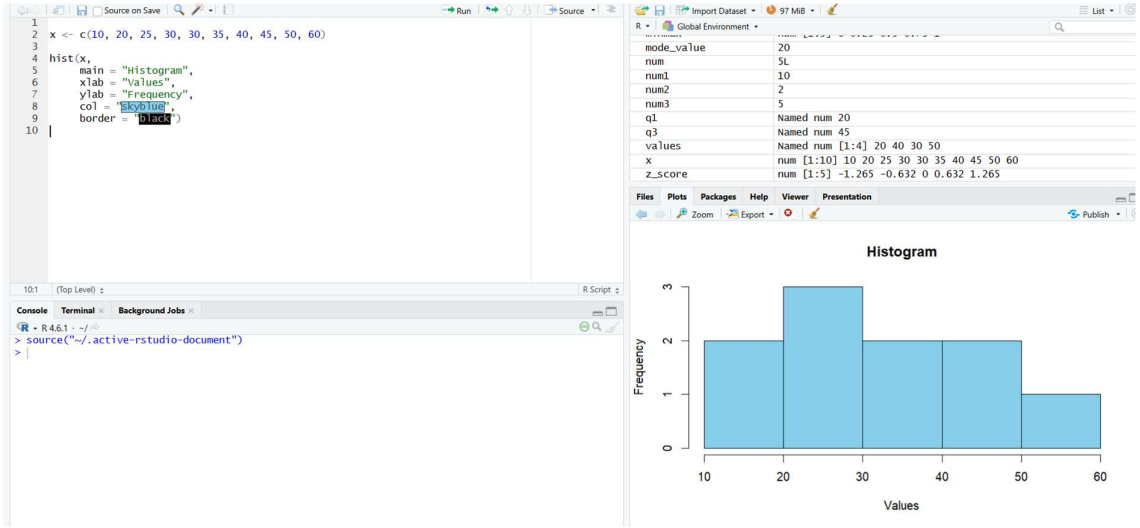
Exp-15:



Result:

Thus, the box plot was executed successfully.

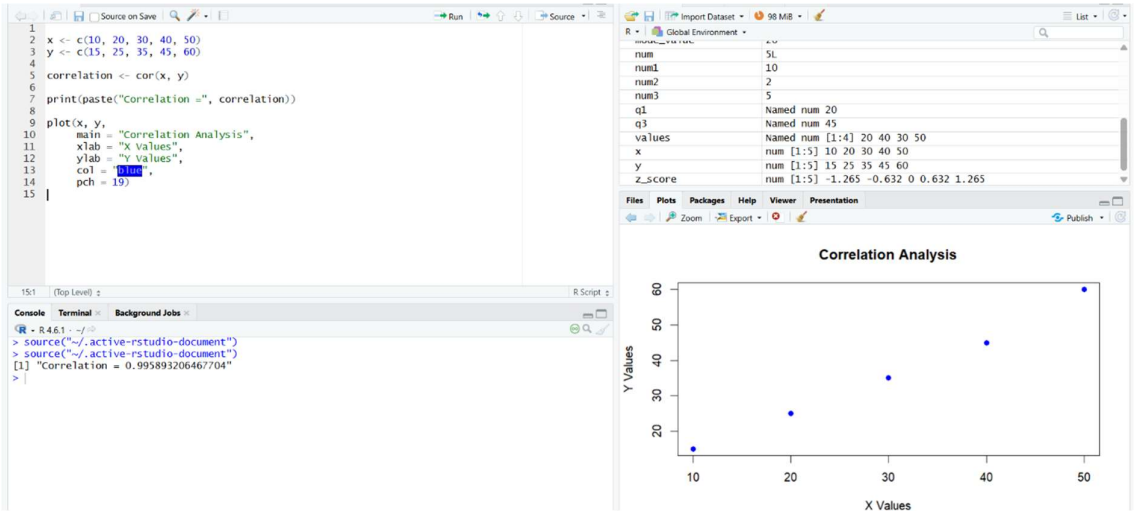
Exp-16:



Result:

Thus, the histogram plot was executed successfully.

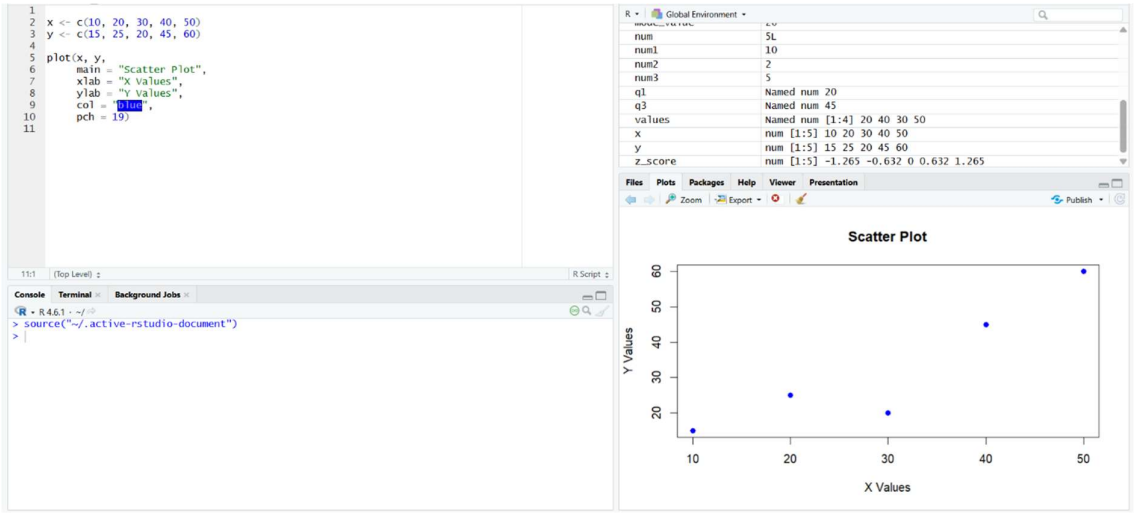
Exp-17:



Result:

Thus, the correlation analysis was executed successfully.

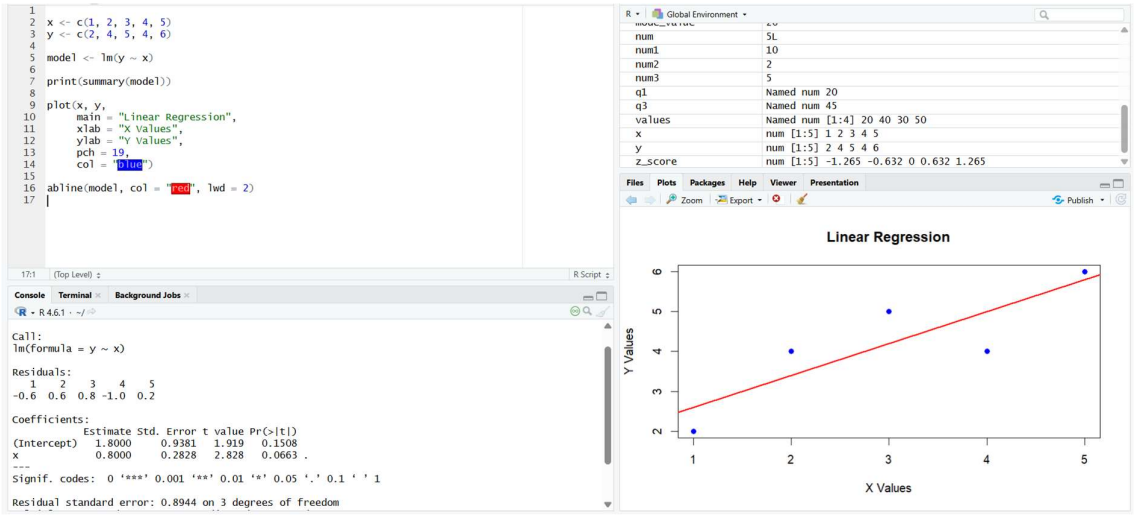
Exp-18:



Result:

Thus, the scatter plot was executed successfully.

Exp-19:



Result:

Thus, the linear regression program was executed successfully.

Exp-20:

```
1
2  y <- c(10, 20, 30, 40, 50)
3  x1 <- c(1, 2, 3, 4, 5)
4  x2 <- c(2, 3, 5, 4, 6)
5
6  model <- lm(y ~ x1 + x2)
7
8  print(summary(model))
9
```

9:1 (Top Level)

Console | Terminal | Background Jobs

R 4.6.1 · ~/

```
Residuals:
         1          2          3          4          5
 1.030e-14 -1.269e-14 -1.040e-15 -1.040e-15  4.474e-15

Coefficients:
             Estimate Std. Error    t value Pr(>|t|)
(Intercept)  1.907e-14  1.697e-14  1.124e+00    0.378
x1           1.000e+01  8.727e-15  1.146e+15   <2e-16 ***
x2          -9.498e-16  8.727e-15 -1.090e-01    0.923
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.203e-14 on 2 degrees of freedom
Multiple R-squared:      1,     Adjusted R-squared:      1
F-statistic: 3.456e+30 on 2 and 2 DF,  p-value: < 2.2e-16
```

Result:

Thus, the Multple regression program was executed successfully.